TUWRC

Start here.

Your team gets one **leader arm** (you move it) and one **follower arm** (it copies you). What the robot learns is up to you. Make sure you can demo it at the end.

The loop

1 Calibrate both arms first. Install with `pip install -e ".[feetech]"`, find ports with `lerobot-find-port`. Middle pose, Enter, move every joint end to end, Enter.

```
lerobot-calibrate --robot.type=so101_follower --robot.port=[FOLLOWER PORT] --robot.id=follower
lerobot-calibrate --teleop.type=so101_leader --teleop.port=[LEADER PORT] --teleop.id=leader
```

2 Teleoperate. Move the leader, the follower copies. Use the same ids in every command.

```
lerobot-teleoperate \
  --robot.type=so101_follower --robot.port=[FOLLOWER PORT] --robot.id=follower \
  --teleop.type=so101_leader --teleop.port=[LEADER PORT] --teleop.id=leader
```

3 Record. Add a camera, log in with `hf auth login`, aim for 50 episodes.

```
lerobot-record [same robot + teleop flags] \
  --dataset.repo_id=$HF_USER/my_task --dataset.num_episodes=50 \
  --dataset.single_task="Put the cube in the box" \
  --robot.cameras="{front: {type: opencv, index_or_path: 0, width: 640, height: 480, fps: 30}}"
```

4 Train. Needs a GPU. Use `mps` on a Mac.

```
lerobot-train --policy.type=act --dataset.repo_id=$HF_USER/my_task \
  --policy.device=cuda --policy.repo_id=$HF_USER/my_policy
```

5 Run it. Same camera, same place as when you recorded.

```
lerobot-rollout --strategy.type=base --policy.path=$HF_USER/my_policy \
  --robot.type=so101_follower --robot.port=[FOLLOWER PORT] --robot.id=follower \
  --robot.cameras="[same as recording]" --task="Put the cube in the box" --duration=60
```

Which model

ACT — start here

One task from your demos. A few hours on one GPU.

`--policy.type=act`

SmolVLA

Follows text instructions. About 4h for 20k steps on an A100.

`--policy.path=lerobot/smolvla_base`

π0.5

Strongest and heaviest. Full finetune wants an 80 GB GPU.

`--policy.type=pi05`

Links

Docs	huggingface.co/docs/lerobot	Tutorial	huggingface.co/docs/lerobot/il_robots
Code	github.com/huggingface/lerobot	SO101	huggingface.co/docs/lerobot/so101
Help	LeRobot Discord		